

Adriana-Stefania Ciupeanu

Postdoctoral Fellow in Mathematical Epidemiology

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Education

PhD	University of Manitoba , Interdisciplinary Individual Studies (Mathematics and Statistics) - Supervisors: Dr. Julien Arino and Dr. Saman Muthukumarana - Thesis: Mathematical and statistical modelling during a pandemic event	Winnipeg, MB, Canada Sept 2019 – Feb 2026
Master of Science	University of Manitoba , Mathematics	Winnipeg, MB, Canada Sept 2016 – Aug 2019
Master of Science	Queen's University , Mathematics and Statistics	Kingston, ON, Canada Sept 2015 – Aug 2016
Bachelor of Science (Honours)	Oxford Brookes University , Mathematics	Oxford, UK Sept 2012 – May 2015

Experience

University of Manitoba , Postdoctoral Fellow - Department of Mathematics - Supervisor: Dr. Julien Arino - Research: Mathematical and statistical modelling of infectious disease dynamics	Winnipeg, MB, Canada May 2026 – Oct 2026 6 months
University of Manitoba , Instructor of Record - Pre-Calculus (MSKL 0100) - Nominated for Extended Education Instructor Excellence Award (2026)	Winnipeg, MB, Canada Sept 2025 – Dec 2025 4 months
University of Manitoba , Lab Coordinator - Calculus 2 (MATH 1700) and Linear Algebra 1 (MATH 1300) - Coordinated laboratory sections across multiple groups - Managed TA duties and scheduling - Oversaw grading quality assurance and designed quiz versions	Winnipeg, MB, Canada Sept 2023 – Apr 2026 2 years 8 months
University of Manitoba , Teaching Assistant - Tutorials across 10+ mathematics courses - Marking from first-year calculus to Mathematics Analysis for Engineering - Guest lectures on graph theory and network science	Winnipeg, MB, Canada 2016 – 2025 9 years

Publications

Mathematical Modeling of the Dynamics of COVID-19 Variants of Concern: Asymptotic and Finite-Time Perspectives Adriana-Stefania Ciupeanu, Marie Varughese, Weston C. Roda, Donglin Han, Qun Cheng, Michael Y. Li pubmed.ncbi.nlm.nih.gov/36097594 (Infectious Disease Modelling)	Nov 2022
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Investigating Air Travel Network Changes in Canada, USA and Europe during COVID-19 Using Open Source Data

June 2023

Adriana-Stefania Ciupeanu, Ashani Wickramasinghe, Saman Muthukumarana, Julien Arino

europepmc.org/article/PPR/PPR591103 (Research Square (Preprint))

Investigation of Global and Local COVID-19 Importation Risks

2020

Julien Arino, Stéphane Portet, Nicolas Bajeux, Adriana-Stefania Ciupeanu
(Report to Public Health Agency of Canada)

Skills

Statistical Methods: Bayesian inference, MCMC, Uncertainty quantification, Sensitivity analysis (PRCC, eFAST)

Mathematical Modelling: ODE compartmental models, CTMC simulations, Metapopulation models

Programming: Python, R, Stan, LaTeX

Data Analysis: Network analysis, Time series analysis, Temporal networks

Awards

Faculty of Graduate Studies Travel Award 2023

Faculty of Graduate Studies Travel Award 2022

Graduate Enhancement of Tri-Agency Stipends Program 2023

International Graduate Student Entrance Scholarship 2020

IMA Prize for Academic Excellence in Mathematics 2015

Service

Mathematics Department, University of Manitoba, Hiring Committee Member Sept 2024 – June 2025

University of Manitoba, Graduate Student Representative Sept 2018 – June 2019

University of Manitoba Graduate Student Association, Committee Member and Department Councillor Sept 2018 – June 2019